# THE DEGREES IN WHICH THE EXT SHEAVES OF A COHERENT MODULE AGAINST A DUALIZING COMPLEX SIT, IN TERMS OF DEPTH

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0ECM (sources/stacks-duality/pinned-original-source.tex line 588).

*Allowed standard prerequisites (editorial).*

- Coherence of the Ext sheaves against a dualizing complex. Same chapter, Lemma labelled lemma-dualizing-schemes, cited at proof lines 631-634. Gives coherence of E^i.
- Stalk computation of the Ext sheaves and the dimension function. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-internal-hom, cited at proof lines 634-638, More on Algebra, Lemma labelled more-algebra-lemma-base-change-RHom part (3), cited at proof lines 638-640, and Same chapter, Lemma labelled lemma-dimension-function-scheme, cited at proof lines 642-644. Reduce the local statements to the algebra.
- The algebraic sitting-in-degrees statement. Dualizing Complexes, Lemma labelled dualizing-lemma-sitting-in-degrees, cited at proof lines 644-646. Supplies parts (1), (2) and (3).

*Source reference.* Original lemma 0A89: sources/stacks-duality/pinned-original-source.tex, source lines 472–472. Same-chapter pointer into the author environment lemma at source line 472; invoked at proof line 609 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 612 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0A6A: sources/stacks-chow/tags.txt, source lines 12838–12838. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 615 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0AWF: sources/stacks-duality/pinned-original-source.tex, source lines 567–567. Same-chapter pointer into the author environment lemma at source line 567; invoked at proof line 625 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0A7U: sources/stacks-chow/tags.txt, source lines 12892–12892. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 627 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $X$ be a locally Noetherian scheme. Let $\omega_X^\bullet$ be a dualizing complex on $X$ with associated dimension function $\delta$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Set $\mathcal{E}^i = \mathcal{E}xt^{-i}_{\mathcal{O}_X}(\mathcal{F}, \omega_X^\bullet)$. Then $\mathcal{E}^i$ is a coherent $\mathcal{O}_X$-module and for $x \in X$ we have*

1. *$\mathcal{E}^i_x$ is nonzero only for $\delta(x) \leq i \leq \delta(x) + \dim(\mathit{Supp}(\mathcal{F}_x))$,*
2. *$\dim(\mathit{Supp}(\mathcal{E}_x^{i+\delta(x)})) \leq i$,*
3. *$\mathit{depth}(\mathcal{F}_x)$ is the smallest integer $i \geq 0$ such that $\mathcal{E}_x^{i+\delta(x)} \neq 0$, and*
4. *we have $x \in \mathit{Supp}(\bigoplus_{j \leq i} \mathcal{E}^j) \Leftrightarrow \mathit{depth}_{\mathcal{O}_{X,x}}(\mathcal{F}_x) + \delta(x) \leq i$.*

*Proof.* Lemma 0A89 tells us that $\mathcal{E}^i$ is coherent. Choosing an affine neighbourhood of $x$ and using Derived Categories of Schemes, Lemma 0A6H and More on Algebra, Lemma 0A6A part (3) we have

$$\mathcal{E}^i_x = \mathcal{E}xt^{-i}_{\mathcal{O}_X}(\mathcal{F}, \omega_X^\bullet)_x = \mathrm{Ext}^{-i}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \omega_{X,x}^\bullet) = \mathrm{Ext}^{\delta(x)-i}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \omega_{X,x}^\bullet[-\delta(x)])$$

By construction of $\delta$ in Lemma 0AWF this reduces parts (1), (2), and (3) to Dualizing Complexes, Lemma 0A7U. Part (4) is a formal consequence of (3) and (1). □